```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

```
sns.set_theme(style='whitegrid')
print("All libraries imported successfully")
```

```
All libraries imported successfully
```

```
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/tit

df = pd.read_csv(url)
display(df.head())
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Tick
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 175
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O 31012
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	3734

```
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])

print("\nColumn Names:")
print(df.columns.tolist())

print("\nDataset Information:")
df.info()
```

```
print("\nStatistical Summary:")
display(df.describe())
```

```
Number of Rows: 891
Number of Columns: 12

Column Names:
['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch',

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

Statistical Summary:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

```
print("Missing Values in Each Column:")
print(df.isnull().sum())
```

```
Missing Values in Each Column:
PassengerId      0
Survived         0
```

```
Pclass          0
Name            0
Sex             0
Age           177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin         687
Embarked        2
dtype: int64
```

```
print("Number of Duplicate Rows:", df.duplicated().sum())
```

```
Number of Duplicate Rows: 0
```

```
print("Survival Counts:")
print(df['Survived'].value_counts())

print("\nPassenger Gender Counts:")
print(df['Sex'].value_counts())

print("\nPassenger Class Counts:")
print(df['Pclass'].value_counts())

print("\nEmbarkation Port Counts:")
print(df['Embarked'].value_counts())
```

```
Survival Counts:
Survived
0    549
1    342
Name: count, dtype: int64

Passenger Gender Counts:
Sex
male      577
female    314
Name: count, dtype: int64

Passenger Class Counts:
Pclass
3    491
1    216
2    184
Name: count, dtype: int64

Embarkation Port Counts:
Embarked
S    644
C    168
Q     77
Name: count, dtype: int64
```

```
survival_rate = df['Survived'].mean() * 100

print("Overall Survival Rate:", round(survival_rate, 2), "%")
```

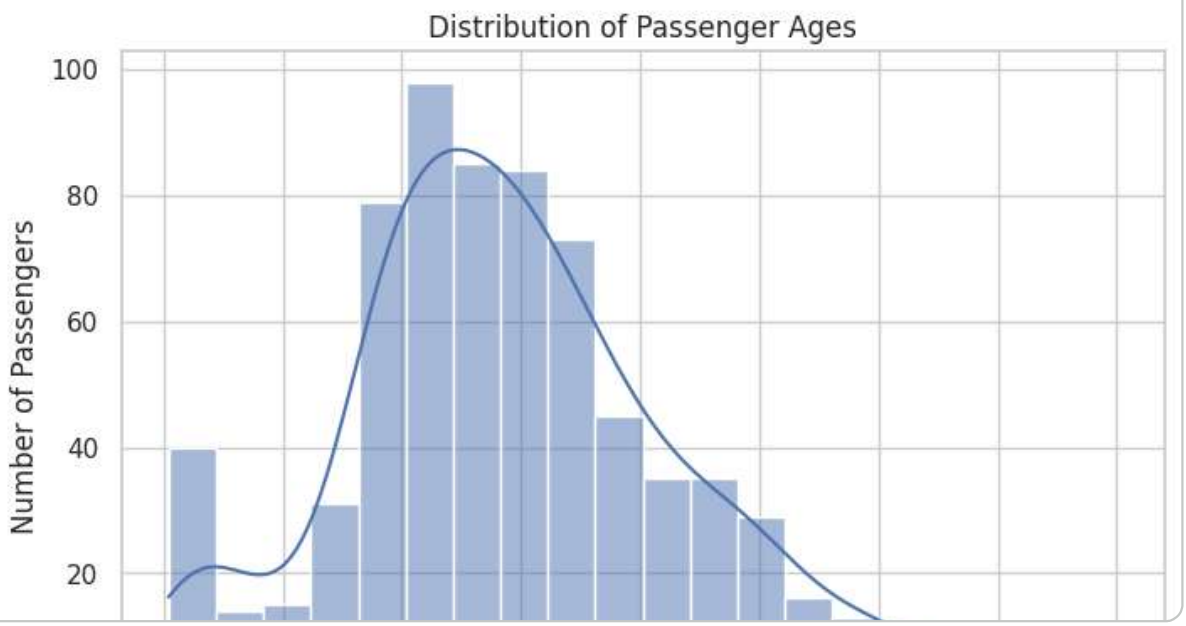
```
Overall Survival Rate: 38.38 %
```

```
plt.figure(figsize=(8, 5))

sns.histplot(data=df, x='Age', bins=20, kde=True)

plt.title('Distribution of Passenger Ages')
plt.xlabel('Age')
plt.ylabel('Number of Passengers')

plt.show()
```




```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x='Fare')

plt.title('Boxplot of Passenger Fares')
plt.xlabel('Fare')

plt.show()
```

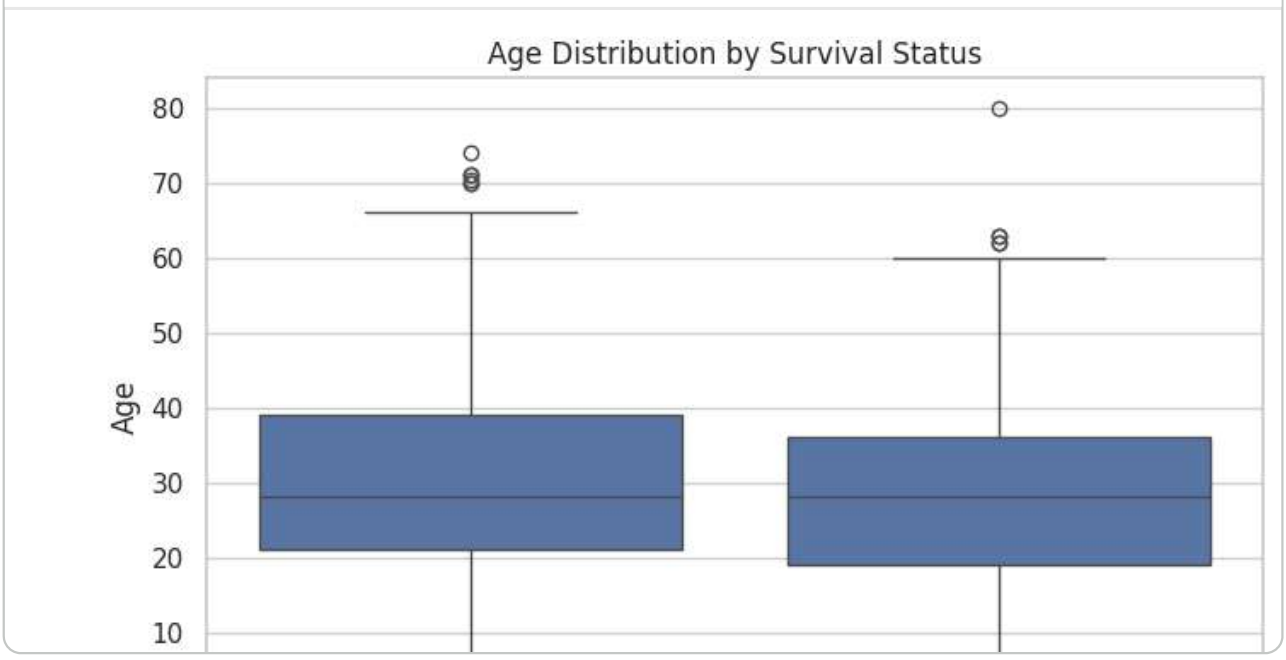



```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x='Survived', y='Age')

plt.title('Age Distribution by Survival Status')
plt.xlabel('Survival Status (0 = No, 1 = Yes)')
plt.ylabel('Age')

plt.show()
```



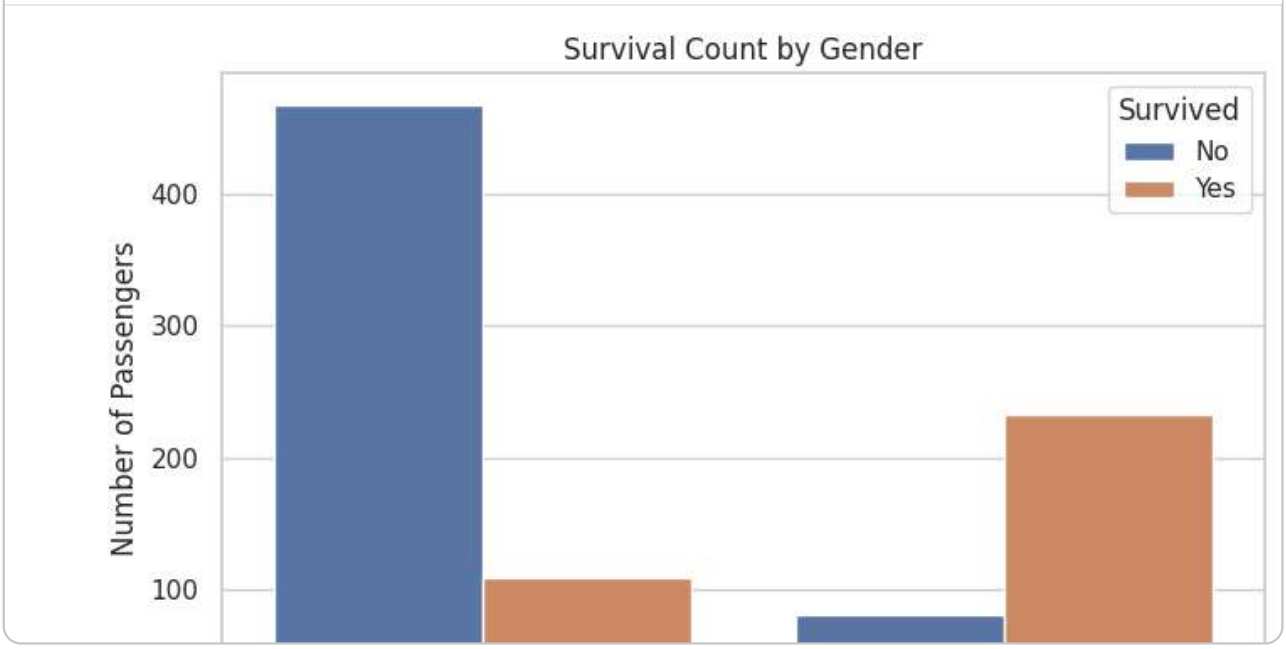

```
plt.figure(figsize=(8, 5))

sns.countplot(data=df, x='Sex', hue='Survived')

plt.title('Survival Count by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')
```

```
plt.legend(title='Survived', labels=['No', 'Yes'])

plt.show()
```



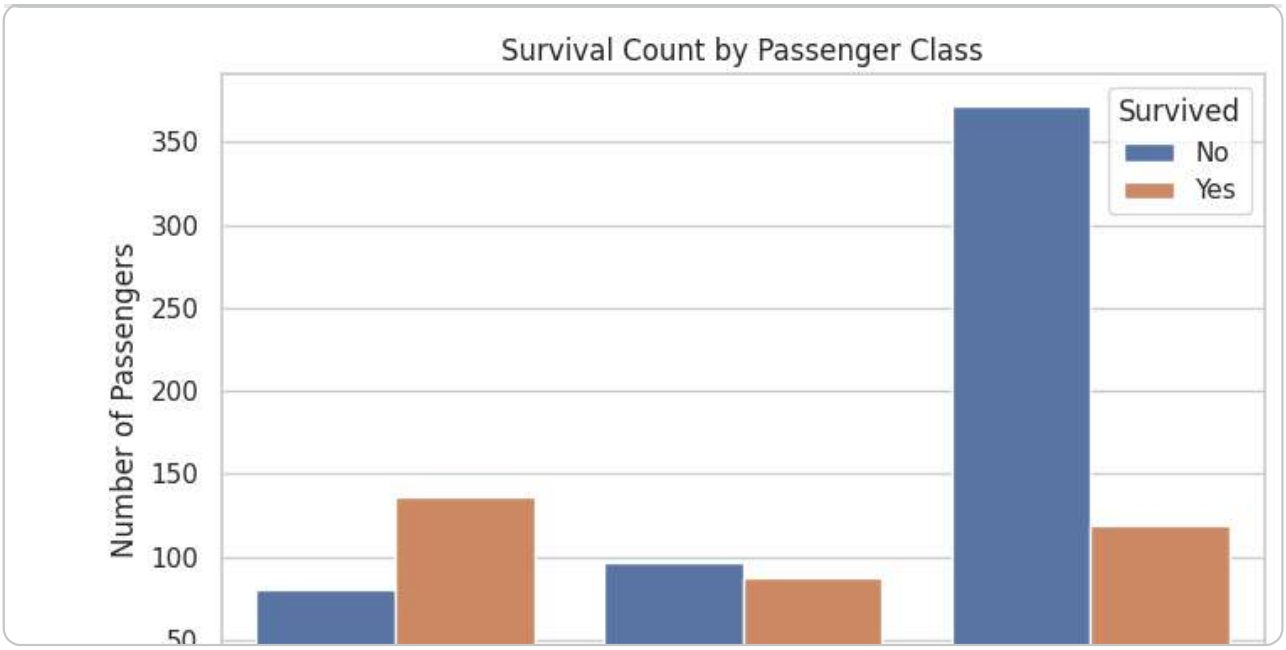
The bar chart shows a clear difference in survival counts between male and female passengers. In the Titanic dataset, female passengers had a higher survival proportion than male passengers. This suggests that gender was associated with survival outcomes.

```
plt.figure(figsize=(8, 5))

sns.countplot(data=df, x='Pclass', hue='Survived')

plt.title('Survival Count by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])

plt.show()
```



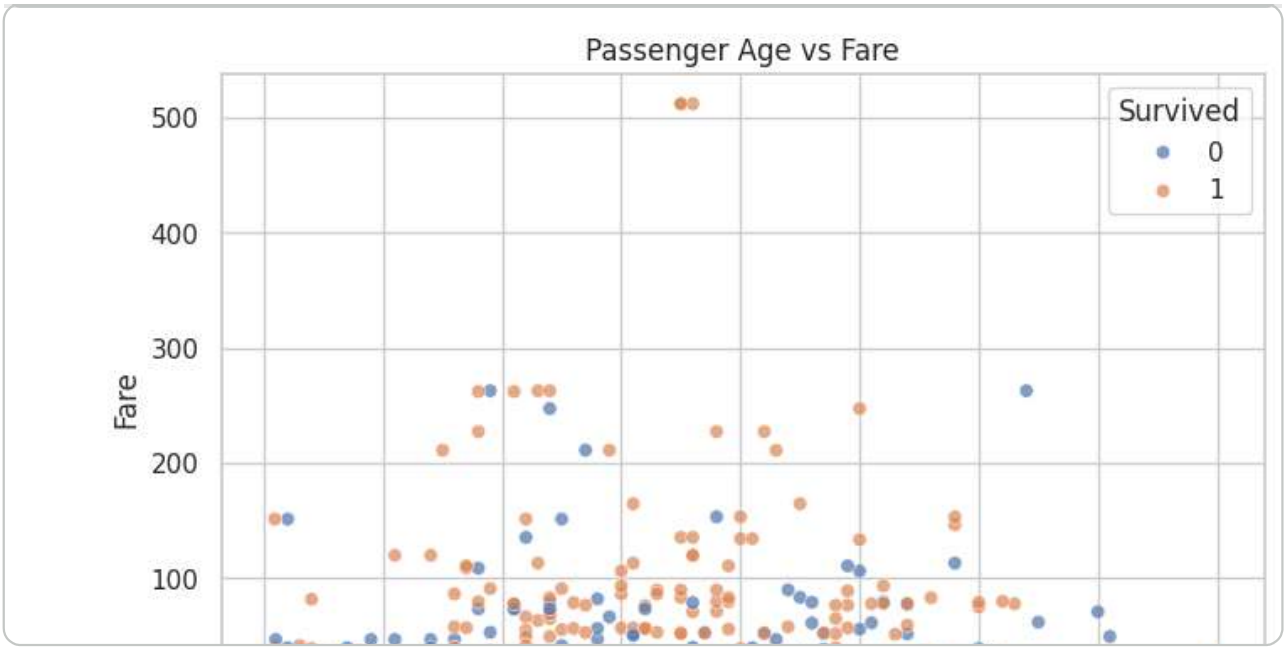
The chart indicates differences in survival outcomes across passenger classes. First-class passengers generally had a higher survival proportion, while third-class passengers experienced lower survival outcomes. Passenger class may be associated with differences in access to evacuation opportunities.

```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x='Age',
    y='Fare',
    hue='Survived',
    alpha=0.7
)

plt.title('Passenger Age vs Fare')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.legend(title='Survived')

plt.show()
```



The scatterplot displays the relationship between passenger age and ticket fare, with colours indicating survival status. Fares vary considerably across passengers of different ages, and some passengers paid much higher fares than others. The chart also helps identify clusters and potential outliers.

```
gender_survival = df.groupby('Sex')['Survived'].mean() * 100

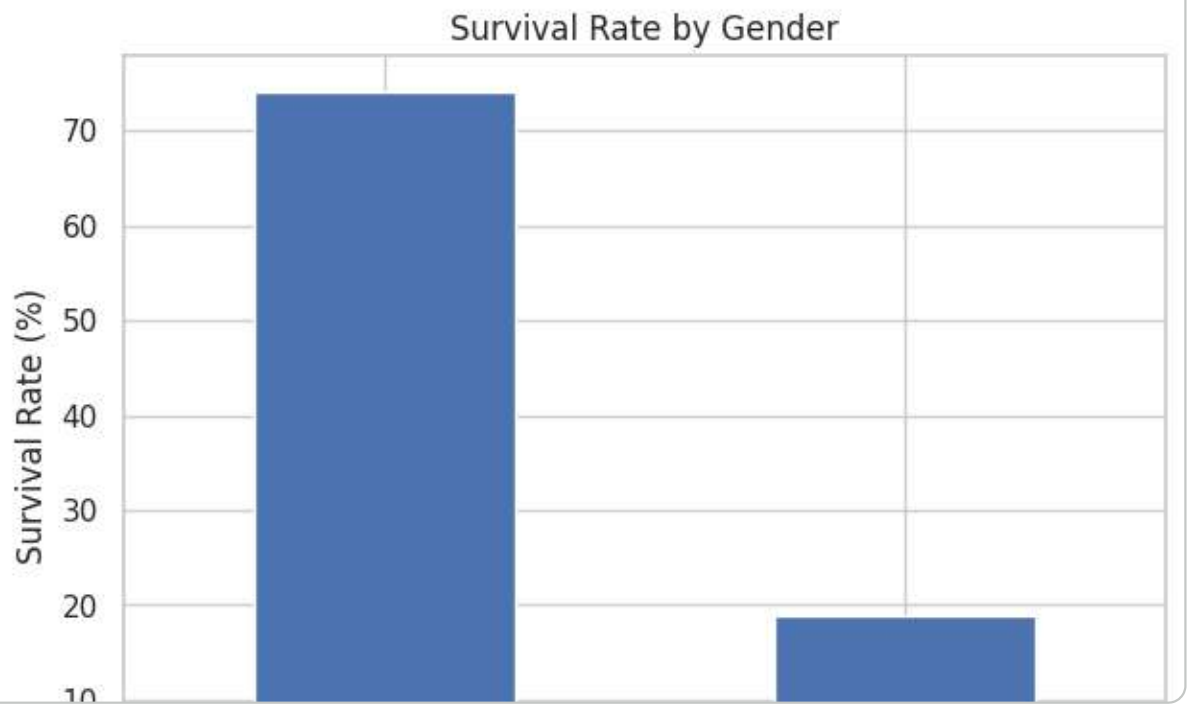
print("Survival Rate by Gender:")
print(gender_survival.round(2))

gender_survival.plot(kind='bar', figsize=(7, 5))

plt.title('Survival Rate by Gender')
plt.xlabel('Gender')
plt.ylabel('Survival Rate (%)')
plt.xticks(rotation=0)

plt.show()
```

```
Survival Rate by Gender:
Sex
female    74.20
male      18.89
Name: Survived, dtype: float64
```



The gender-wise survival rate comparison shows that female passengers had a substantially higher survival rate than male passengers in this dataset. This is a descriptive finding and does not by itself prove causation.

```
numeric_df = df.select_dtypes(include='number')

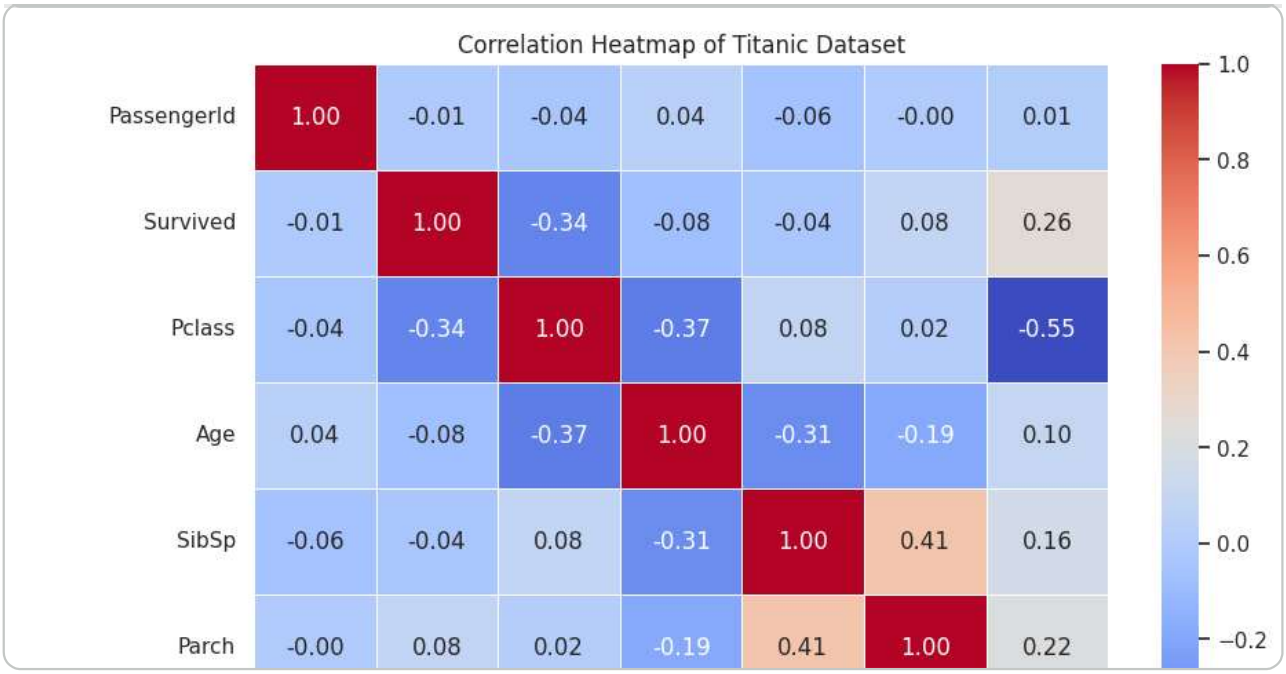
correlation_matrix = numeric_df.corr()

plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap='coolwarm',
    fmt='.2f',
    linewidths=0.5
)

plt.title('Correlation Heatmap of Titanic Dataset')

plt.show()
```



Observation – Correlation Heatmap

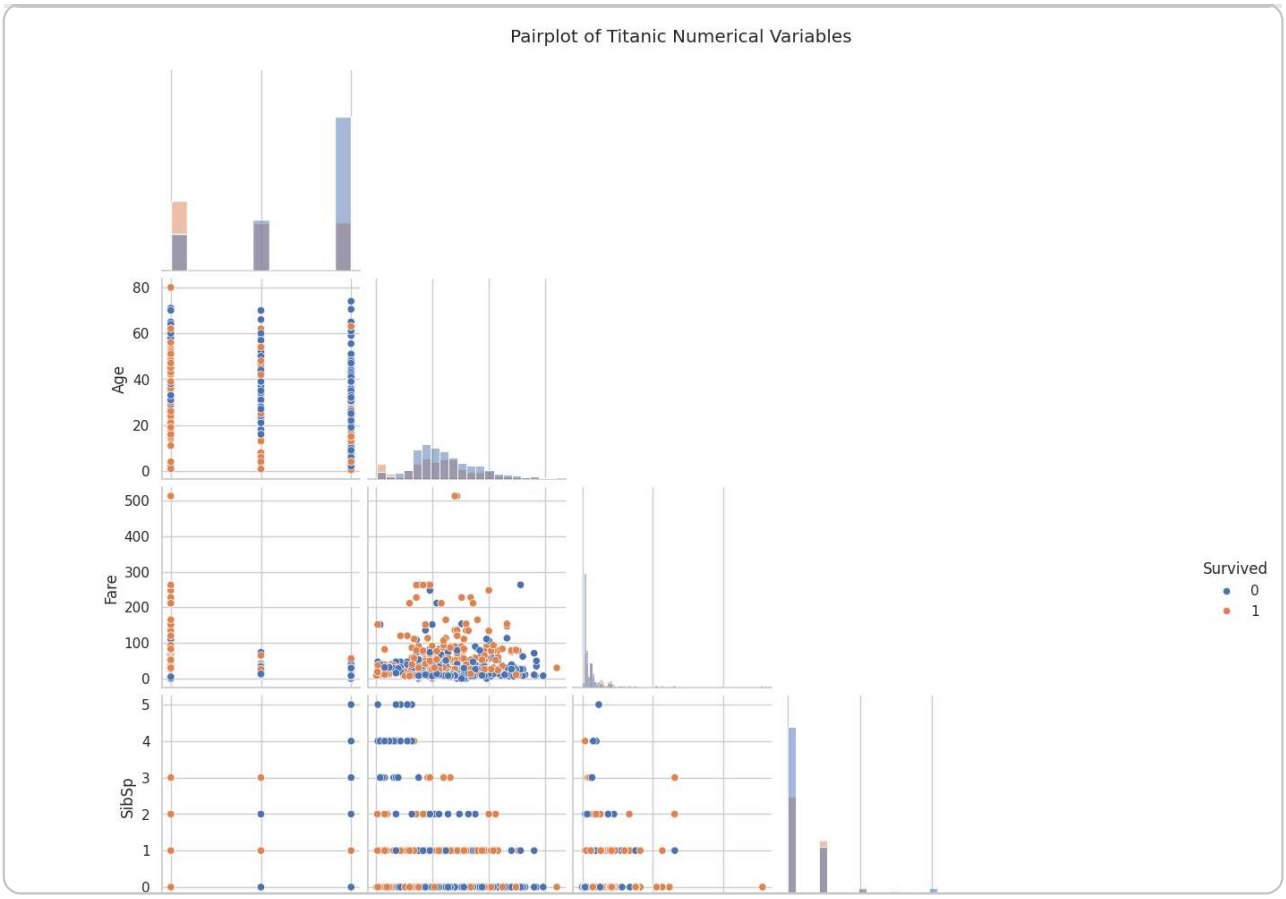
The correlation heatmap summarizes the linear relationships between numerical variables. Passenger class and fare may show a negative correlation because lower class numbers represent higher passenger classes, which generally have higher fares. Correlation does not establish causation, and a low correlation does not rule out a nonlinear relationship.

```
pairplot_data = df[
    ['Survived', 'Pclass', 'Age', 'Fare', 'SibSp', 'Parch']
].dropna()

sns.pairplot(
    pairplot_data,
    hue='Survived',
    diag_kind='hist',
    corner=True
)

plt.suptitle(
    'Pairplot of Titanic Numerical Variables',
    y=1.02
)

plt.show()
```



Close

```
**Observation – Pairplot**

The pairplot provides an
overview of distributions
and pairwise relationships
among passenger survival,
class, age, fare, siblings
or spouses, and parents or
children aboard. The colour
groups help compare
survivors and
non-survivors. Some
relationships may show
overlapping patterns,
indicating that no single
numerical variable fully
explains survival outcomes.
```

Observation – Pairplot

The pairplot provides an overview of distributions and pairwise relationships among passenger survival, class, age, fare, siblings or spouses, and parents or children aboard. The colour groups help compare survivors and non-survivors. Some relationships may show overlapping patterns, indicating that no single numerical variable fully explains survival outcomes.

```
from statsmodels.stats.outliers_influence import variance

vif_data = df[
    ['Pclass', 'Age', 'SibSp', 'Parch', 'Fare']
].copy()

# Fill missing numerical values with their column median
vif_data = vif_data.fillna(vif_data.median())

vif_results = pd.DataFrame()
```

```
vif_results['Feature'] = vif_data.columns
vif_results['VIF'] = [
    variance_inflation_factor(vif_data.values, i)
    for i in range(vif_data.shape[1])
]

display(vif_results.round(2))
```

1 to 5 of 5 entries Filter

index	Feature	VIF
0	Pclass	1.66
1	Age	1.2
2	SibSp	1.28
3	Parch	1.26
4	Fare	1.59

Show 25 per page
Like what you see? Visit the data table notebook to learn more about interactive tables.

```
print("EXPLORATORY DATA ANALYSIS - KEY FINDINGS")
print("=" * 45)

# Overall survival rate
survival_rate = df['Survived'].mean() * 100
print(f"\n1. Overall Survival Rate: {survival_rate:.2f}%

# Survival rate by gender
gender_rate = df.groupby('Sex')['Survived'].mean() * 100
print("\n2. Survival Rate by Gender:")
print(gender_rate.round(2))

# Survival rate by passenger class
class_rate = df.groupby('Pclass')['Survived'].mean() * 1
print("\n3. Survival Rate by Passenger Class:")
print(class_rate.round(2))

# Average fare by passenger class
fare_by_class = df.groupby('Pclass')['Fare'].mean()
print("\n4. Average Fare by Passenger Class:")
print(fare_by_class.round(2))

# Missing values
print("\n5. Columns with Missing Values:")
print(df.isnull().sum()[df.isnull().sum() > 0])

# Age statistics
print("\n6. Passenger Age Statistics:")
print(df['Age'].describe().round(2))
```

```
EXPLORATORY DATA ANALYSIS - KEY FINDINGS
=============================================
```

```
1. Overall Survival Rate: 38.38%

2. Survival Rate by Gender:
Sex
female    74.20
male      18.89
Name: Survived, dtype: float64

3. Survival Rate by Passenger Class:
Pclass
1    62.96
2    47.28
3    24.24
Name: Survived, dtype: float64

4. Average Fare by Passenger Class:
Pclass
```